

AMENDMENT NO. 1 JANUARY 2021

TO

IS 9438 : 2018 PERFORMANCE REQUIREMENTS AND METHODS OF TESTS FOR WHEELS/ RIMS FOR TRUCKS AND BUSES

(First Revision)

(Foreword, para 1, para 2 and para 3) — Substitute the following for the existing paras:

“This Indian Standard (First Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Automotive Tyres, Tubes and Rim Sectional Committee had been approved by the Transport Engineering Division Council.’

This standard was first published in 1980. This revision has been undertaken to align this standard with international practices. This standard is in agreement with the international Standard ISO 3894 : 2015 ‘Road vehicles — Wheels/ rims for commercial vehicles — Test methods’, issued by the International Organization for Standardization. It also includes the performance requirements apart from test methods. In this revision the aluminium wheels have also been included and accelerated test factors for cornering fatigue test have been specified.

In the formulation of this standard, assistance has also been derived from SAEJ267_201411 ‘Wheels/ Rims — Truck and Bus – Performance Requirements and Test Procedures for Radial and Cornering Fatigue’, issued by the Society of Automotive Engineers (Inc), USA and ISO 3894 : 2015 ‘Road vehicles — Wheels/rims for commercial vehicles — Test methods.’”

Price Group 3

(Page 1, Title) — Substitute the following for the existing title:

Indian Standard

**PERFORMANCE REQUIREMENTS
AND METHODS OF TESTS FOR
WHEELS/ RIMS FOR TRUCKS
AND BUSES**

(*First Revision*)

(Page 1, clause 1) — Substitute the following for the existing clause:

‘1 SCOPE

This standard specifies three laboratory test methods and performance requirements for evaluating certain essential Fatigue strength characteristics of disc wheels, spoke wheels and demountable rims intended for, but not limited to road use on commercial vehicles of categories N2, N3, M2, M3 and Trailers of T3, T4 and T5 categories as defined in IS 14272.

The test methods are:

- a) Dynamic cornering fatigue test (for disc wheel rims),
- b) Dynamic radial fatigue test (for disc wheel rims and wheel with demountable rims),
- c) Dynamic cornering fatigue (for wheels with demountable rims).’

(Page 1, clause 3.1) — Substitute the following for the existing clause:

‘3.1 Wheel Rim — The Wheel rim is the outer edge of a wheel, holding the tyre. It makes up the outer circular design of the wheel on which the inside edge of the tyre is mounted on vehicles such as automobiles to agreements based on this standard.’

(Page 1, clause 3.2) — Substitute the following for the existing clause:

‘**3.2 Static Loaded Radius** — The measurement in metres from the wheel axle centerline to the ground when the tyre is properly inflated corresponding to the rated load.’

(Page 1, clause 3.3.1) — Substitute the following for the existing clause:

‘**3.3.1 Inset** — The offset is the distance between the mounting face of the disc and the rim centerline. This distance is called inset when the mounting face is outside of the rim centerline.’

(Page 1, clause 3.3.2) — Substitute the following for the existing clause:

‘**3.3.2 Outset** — The offset is the distance between the mounting face of the disc and the rim centerline. This distance is called outset when the mounting face is inside of the rim centerline.’

(Page 1, clause 3.4) — Substitute the following for the existing clause:

‘**3.4 Drop Centre Rim** — The centre section of the rim being lower than the two outer edges, this allows the bead of the tyre to be pushed into the low area on one side while the other side is pulled over off the flange.’

(Page 1, clause 4) — Substitute the following for the existing clause:

‘4 GENERAL

Only fully processed new wheels/rims which are representative of wheels/rims intended for the vehicle shall be used for the tests. No wheel/rim shall be used for more than one test.’

(Page 1, clause 5.1.1) — Substitute the following for the existing clause:

‘5.1.1 Equipment

The test machine shall have a driven rotatable device whereby either the wheel rotates under the influence of a stationary bending moment or the wheel is stationary and is subject to a rotating bending moment [see Fig. 1 A and Fig. 1B].’

(Page 1, clause 5.1.2.1) — Substitute the following for the existing clause:

‘5.1.2.1 Preparation

Clamp the rim of the wheel securely to the test machine with the help of suitable adaptors/jigs in accordance with Fig. 1 A or Fig. 1 B. The adaptor face of the test machine shall have equivalent mounting systems to those used on the vehicle. The mating surface of the test adaptor and wheel shall be free of excessive scoring and deformation, and excessive build-up of paint, dirt or foreign matter.

Attach the loading arm (moment arm) and adaptor assembly to the mounting surface of the wheel using studs or bolts, and nuts which are in good condition, lubricated or non-lubricated in accordance with the intended vehicle application (as specified by the vehicle manufacturer), and are representative of those used on the vehicle. Assemble and tighten the wheel fixing nuts and bolts at the beginning of the test using the procedure specified by the vehicle or wheel manufacturer.

Wheel bolts or nuts may be retorqued during the test.’

(Page 2, clause 5.1.3) — Substitute the following for the existing clause:

‘5.1.3 Bending Moment Application

To impart a bending moment to the wheel, apply a force parallel to the plane of the wheel mounting surface at a specified distance with the loading arm (moment arm) as shown in Fig. 1 A or Fig. 1 B.

Maintain the bending moment within the accuracy of ± 5 percent of the calculated value.’

(Page 2, clause 5.1.4) — Substitute the following for the existing clause:

‘5.1.4 Bending Moment Determination

Determine the bending moment M (force $\times$ loading/moment arm), given in Newton-metres, by the following formula:

$$M = (\mu \cdot R + d) F_v \times S$$

where

μ = assumed coefficient of friction developed between tyre and road (see Table 1);

R = static loaded radius, in metres, of the largest tyre to be used on the wheel as specified by the vehicle manufacturer;

d = inset or outset (positive for inset; negative for outset), of the wheel, in metres;

If the wheel may be used as an inset and outset wheel, then the inset value shall be used;

F_v = load rating of the wheel, in Newton, as specified by vehicle or wheel manufacturer; and

S = accelerated test factor (*see* Table 1).

(Page 3, Fig. 1, Key) — Substitute the following for the existing clause:

‘Key

- a Test load
- b Rim centre plane
- c Inset, d
- d Diameter
- e Loading/Moment arm, l'

[Page 4, clause 5.1.6 (c)] — Substitute the following for the existing clause:

‘c) The wheel shall withstand a minimum no. of performance cycles without failure as mentioned at Annex A (*see* Table 1).’

(Page 4, clause 5.2.1) — Substitute the following for the existing clause:

‘5.2.1 Equipment

The test machine shall be equipped with a means of imparting constant radial load as the wheel rotates. There are many means of imparting radial loads. The suggested equipment incorporates a driven rotatable drum set which presents a smooth surface wider than the loaded test tyre section width. The recommended minimum external diameter of the drum is 1 700 mm $\pm$ 1 percent. The test wheel (single application) and tyre fixture shall provide loading normal to the surface of the external surface and inline radially with the centre of the test wheel and drum. The axes of the drum and test wheel shall be parallel.’

(Page 4, clause 5.2.1.1) — Substitute the following for the existing clause:

‘5.2.1.1 Procedure

Tyres selected for this test shall meet the load rating F_v of the wheel or be representative of the maximum load capacity tyre specified by the vehicle or wheel manufacturer, whichever is greater. For disc wheels, the test adaptor shall be representative of production hubs using bolts or studs, and nuts which are in good condition, lubricated or non-lubricated in accordance with the intended vehicle application (as specified by the vehicle manufacturer), and are representative of those used on the vehicle. For demountable rims, the test adaptor shall be representative of production spoke wheels using bolts or studs and nuts and clamps which are in good condition, lubricated or non-lubricated in accordance with the intended vehicle application(as specified by the vehicle manufacturer), and are representative of those used on the vehicle. Torque the wheel nuts to the torque limit specified by the vehicle or wheel manufacturer for the stud size and type of nut used. Check nut torque values and reset them periodically during the course of the test in order to compensate for the ‘wearing-in’ of mating surfaces of nut and bolt holes.

The test load inflation pressure is based on wheel/rim rating. Test inflation pressures shown below are for information only.’

(Page 4, clause 5.2.2) — Substitute the following for the existing clause:

‘5.2.2 Determine the radial load F_r in Newton, by the following formula:

$$F_r = F_v \times K$$

where

F_v = load rating of wheel in Newton, as specified by the vehicle or wheel manufacturer; and

K = accelerated test load factor (see Table 2).’

(Page 4, clause 5.2.3) — Substitute the following for the existing clause:

‘5.2.3 Minimum No. of Performance Cycles to be Completed

The wheel rim shall withstand minimum no. of performance cycles without failure as specified in Table 2, prior to test termination.’

(Page 4, clause 5.2.4) — Substitute the following for the existing clause:

‘5.2.4 Test Termination

The test shall be terminated in any of the three following circumstances:

- a) Inability of wheel rim to sustain the calculated test radial load; or
- b) Propagation of a crack or cracks existing prior to test or new visible stress-caused cracks penetrating through a section of the wheel; or
- c) The wheel shall withstand a minimum load cycles without failure as mentioned at Annex A (see Table 2).’

(Page 4, clause 5.3.1) — Substitute the following for the existing clause:

‘5.3.1 Equipment

The test machine shall have a driven rotatable device whereby either the wheel rotates under the influence of a stationary bending moment or the wheel is stationary and is subjected to a rotating bending moment(see Fig. 3).’

(Page 4, clause 5.3.2.1) — Substitute the following for the existing clause:

‘5.3.2.1 Preparation

Clamp the wheel to the test machine with the help of suitable adaptors/jigs. To ensure this, assemble and tighten the wheel fixing at the beginning of the test using the procedures specified by the vehicle or wheel manufacturer. Attach a rigid loading/moment arm shaft with a test hub adaptor to the hub wheel. The mating surfaces of the test adaptor and wheel shall be free of excessive build-up of paint, dirt or foreign material. Use studs, nuts and clamps which are in good condition, lubricated or non-lubricated in accordance with the intended vehicle application (as specified by the vehicle manufacturer), and are representative of those used on the vehicle or specified for the rim.

If the wheel application is always used with a brake drum, the wheel may be tested with a brake drum attached. If the wheel application is ever to be used without a brake drum, the wheel shall be tested without a brake drum attached.’

(Page 5, clause 5.3.4) — Substitute the following for the existing clause:

‘5.3.4 Bending Moment Determination

Determine the bending moment M (force $\times$ loading/moment arm), in Newton-metres, by the following formula:

$$M = \mu \times R \times F_v \times S$$

where

μ = assumed coefficient of friction developed between tyre and road (*see* Table 3);

R = static loaded radius, in metres, of the largest tyre to be used on the wheel as specified by the vehicle or wheel manufacturer;

F_v = load rating of the wheel, in Newton, as specified by the vehicle or wheel manufacturer; and

S = the accelerated test factor (*see* Table 3).’

(Page 5, clause 5.3.5) — Substitute the following for the existing clause:

‘5.3.5 Minimum No. of Performance Cycles to be Completed

The wheel rim shall withstand minimum number of performance cycles without failure as specified in Table 3, prior to test termination.’

[Page 5, clause 5.3.6 (b)] — Substitute the following for the existing clause:

‘b) Propagation of a crack or cracks existing prior to test visible stress-caused cracks penetrating through a section of the wheel.’

[Page 5, clause 5.3.6 (c)] — Substitute the following for the existing clause:

‘c) The wheel shall withstand a minimum number of performance cycles without failure as mentioned at Annex A (*see* Table 3).’

(Page 6, Fig. 3B, Key) — Substitute the following for the existing:

‘Key

a Loading/Moment arm

b Test load

- c Bevel width
- d $b/2$
- e Centre plane of spacer used for specified assembly'

(Page 7, clause 7) — Substitute the following for the existing clause:

‘7 MARKING

7.1 Wheel rims shall be durably and legibly marked with the following:

- a) Size designation as per IS 10694 (Part 1),
- b) Name or trade-mark of the wheel rim manufacturer,
- c) Date of manufacturing preferably in the form of year and month or year and quarter of the year (for example 2009 indicates Sept 2020; 20 IV indicates fourth quarter of 2020); and
- d) Rim bead seat contour type, if applicable,

7.2 The markings shall be made as per 4 of IS 10694 (Part 1).

8 BIS CERTIFICATION MARKING

8.1 The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the products may be marked with the standard mark.

8.2 The application for certification shall contain at least the technical information as specified in Annex B.'

[Page 7, Table 1, Sl No. i), col (5)] — Substitute the following for the existing matter:

‘Inset below 100 and all outset’

(Page 8, Table 2, Title) — Substitute the following for the existing title :

‘Table 2: Test Factors for Dynamic Radial Fatigue Test, Loading Method and Minimum No. of Performance Cycles

(Clauses 5.2.2 and 5.2.4)’

[Page 8, Table 2, Sl No. i), col (4)] — Substitute the following for the existing matter:

‘13, 14, 15, 16, 17 (5° drop centre rims and 5° semi drop centre rims)’

[Page 8, Table 2, Sl No. ii), col (4)] — Substitute the following for the existing matter:

‘15, 17, 18, 20, 22, 24, 17.5, 19.5, 22.5, 24.5’

(Page 8, Table 3) — Substitute the following for the existing table:

**Table 3 Test Load Factors for Wheels with Demountable Rims —
Dynamic Cornering Fatigue Test**
(Clauses 5.3.4 and 5.3.5)

Sl No.	Material	Accelerated test factor ‘S’	Coefficient of friction ‘μ’	Min No. of Performance Cycles to be Completed
(1)	(2)	(3)	(4)	(5)
i)	Ferrous all	1.9	0.7	18 000
		1.5		30 000

(Page 9, Annex B) — Substitute the following for the existing:

ANNEX B

(Clauses 6.2 and 8.2)

INFORMATION TO BE SUBMITTED FOR CERTIFICATION OF WHEEL RIMS

B-1 Following information shall be submitted for certification of wheel rims:

- 1 Manufacturer’s Name and Address
- 2 Telephone No.

- 3 Fax No.
- 4 E-mail Address
- 5 Contact Person
- 6 Wheel Rim Designation
- 7 Wheel Rim category (*Disc wheel/ Wheel with demountable rim*)
- 8 Wheel Type
- 8.1 Material [Light Alloy (Aluminium, Magnesium, etc)/Ferrous (Steel etc.)]
- 8.2 Wheel Mounting holes (Nos.)
- 8.3 Maximum (Front/ Rear) load capacity (kN)
- 8.4 Recommended inflation Service Pressure (kPa)
- 9 Inset/Outset (mm)
- 10 Dynamic loaded radius (mm)
- 10.1 Static loaded radius (mm)
- 11 Tightening torque for studs and nuts
- 11.1 Size of Nut/Stud
- 12 Detail of necessary accessories (if applicable)
- 12.1 Suitable for tubeless tyre mounting
- 13 Drawing, sufficiently detailed to permit identification of the type. They shall also show the position intended for the approval mark and for the wheel markings